A Clarification of the Derivation in Zhang and Boos (2023)

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This note concerns a recent preprint by Nicolas and Hotz (2025), which revisits the zero-surface-humidity temperature bound proposed in Zhang and Boos (2023). We appreciate their careful attention to this problem and showing that the steep temperature lapse rate within the surface layer contributes to the violation of the dry adiabatic bound in our testing of theory against observations. I also acknowledge that some of their claims were revised in the final published version. Nonetheless, the original claim of an inconsistency in the derivation is based on an incorrect interpretation of a key assumption in Zhang and Boos (2023), namely that the mid-to-upper troposphere is approximately on a moist adiabat during dry heat events. A more detailed explanation is provided on the following pages.

Additionally, Zhang and Boos (2023) also introduced an adjusted bound, obtained by subtracting the zero-humidity bound with a climatological minimum summer surface humidity $q_{\min}$ (scaled by L_v/c_p). This adjusted bound is more practically relevant in interpretaing regional observations (see Figs. 3d–f and Section "Insight into Recent Heatwaves") and demonstrates the utility of a moist-convective scaling for dry heatwaves. Although this adjusted bound is similar to the moist-convective bound in Nicolas and Hotz (2025), it was not discussed in the preprint, but was mentioned in the final revised version.

Finally, I acknowledge that some confusion may have arisen from the application of the theory to observations. In Zhang and Boos (2023), the 500-hPa level was used to represent the nearly moist-adiabatic mid-to-upper troposphere. However, over elevated topography, 500 hPa can reside within the dry-adiabatic boundary layer. A corrected version of Fig. 3a from Zhang and Boos (2023), excluding such locations, is shown in Fig. 2b.

The excluded cases are those emphasized by Nicolas and Hotz (2025). They are therefore largely orthogonal to the mechanism described in Zhang and Boos (2023) and do not provide evidence for or against its validity.

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Detailed explanation

Zhang and Boos (2023) formulated the question of extreme heat as seeking an upper bound. In particular, Zhang and Boos (2023) presents a top-down control on surface temperature from an approximately moist-adiabatic free troposphere in midlatitude summer, a mechanism that has received relatively little attention in the context of dry midlatitude heatwaves. There are multiple perspectives from which a surface temperature bound can be derived, including the textbook dry convective neutrality perspective and the surface energy budget perspective. In contrast, the temperature bounds in Zhang and Boos (2023) are intended specifically to represent the top-down control of the moist-adiabatic free troposphere. To this end, a key assumption stated in the Physical Mechanism and Theory section is that MSE^*_{500} serves as a proxy for the approximately uniform MSE^* above the LCL, which gives

$$\mathrm{MSE}^*_{500}(T_{500}) \approx \mathrm{MSE}^*_{\mathrm{tp}}, \tag{1}$$

where "tp" denotes the tropopause. Neglecting saturation vapor pressure at the tropopause, this implies

$$c_pT_{500} + L_vq^*_{500} + gz_{500} \approx c_pT_{tp} + gz_{tp}. \tag{2}$$

With the bulk stability constraint $\mathrm{MSE}_s \leq \mathrm{MSE}^*_{500}$, we obtain

$$c_pT_s + gz_s \leq c_pT_{500} + L_vq^*_{500}(T_{500}) + gz_{500}(T_{500}) - L_vq_s. \tag{3}$$

Obtaining a uniform bound. Zhang and Boos (2023) has made justifiable approximations to z_{500} in order to write the first three terms as functions of a single variable (T_{500}). Still, the variety of q_s across regions and events prevents a uniform bound. The fewer unknowns, the more useful a theory. We then amplify the right-hand side—that is, by minimizing q_s—thereby preserving the inequality and obtaining a uniform bound independent of q_s. Apparently $q_s \geq 0$, therefore setting $q_s = 0$ leads to

$$c_pT_s + gz_s \leq c_pT_{500} + L_vq^*_{500}(T_{500}) + gz_{500}(T_{500}). \tag{4}$$

This yields the zero-humidity bound in Zhang and Boos (2023).

The zero-humidity bound is perhaps best viewed as the limiting case of the adjusted bound also proposed in Zhang and Boos (2023). Also following the approach of minimizing q_s to a uniform value, we minimize q_s in Eq. (3) to a climatological summertime minimum humidity q_{min} which gives

$$c_pT_s + gz_s \leq c_pT_{500} + L_vq^*_{500}(T_{500}) + gz_{500}(T_{500}) - L_vq_{\mathrm{min}}. \tag{5}$$

Because q_{min} varies across regions, this adjusted bound can only be demonstrated region-

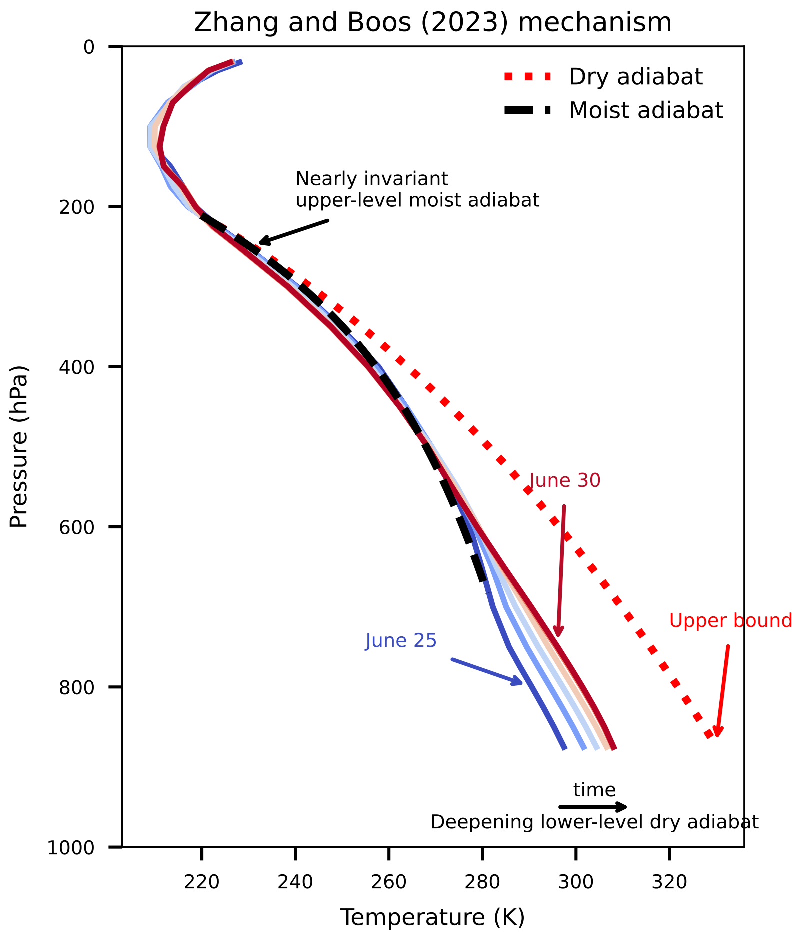

Figure 1: Schematic of the top-down control mechanism described in Zhang and Boos (2023), constructed by annotating temperature profiles from the Pacific Northwest heatwave of 2021.

ally. This is shown for Western Europe, Russia, and the Pacific Northwest (Figs. 3e–f).

Physical picture. Zhang and Boos (2023) describes the following progression: the free troposphere warms, creating an inversion; the boundary layer then warms to reduce this inversion, bringing the temperature profile closer to an adiabat, with a moist adiabat aloft connected to a dry adiabat below; if the anticyclone persists, continued surface drying deepens the dry-adiabatic portion of the profile, which progressively erodes into the moist-adiabatic portion and allows further surface warming until the anticyclone moves away (see Figure 1 for a schematic generated by marking up temperature profiles of an observed event). The progression of vertical temperature profiles for three record heatwaves in three aforementioned regions (Fig. S6) roughly aligns with this conceptual picture. The zero-humidity bound would then correspond to the limiting case in which this process continues until the moist layer vanishes and the column becomes entirely dry-adiabatic (red dotted profile in Fig. 1). This limit may not be realistic, but it is physically consistent.

Connection to textbook dry adiabats. Comparing to a dry adiabat connecting the surface to 500 hPa, which gives

$$c_p T_s + g z_s \leq c_p T_{500} + g z_{500}, \tag{6}$$

the zero-humidity bound in Zhang and Boos (2023) yields

$$c_p T_s + g z_s \leq c_p T_{\mathrm{tp}} + g z_{\mathrm{tp}}, \tag{7}$$

corresponding to a dry adiabat connecting the surface to the tropopause. This distinction follows from Eq. (1) which reflects the assumption that MSE^*_{500} serves as a proxy for the approximately uniform MSE^* above the LCL.

Testing theory with observations The confusion arises from our inclusion of data points for which the 500-hPa level lies within the dry adiabatic portion of the temperature profile during summer months. I therefore exclude such locations, which account for approximately 19% of the total land area between 45°N and 60°N.

The original Figure 3a in Zhang and Boos (2023) is shown alongside the corrected version, together with their difference, in Fig. 2. The excluded regions are highlighted in Fig. 3 and are located primarily over elevated topography and dry deserts.

Excluding these regions produces a downward shift in the distribution, and the upper edge lies farther from the theoretical upper bound. This is not unexpected. The upper bound is derived under highly idealized assumptions and is not intended to be realized exactly in observations. Moreover, the distribution near the bound is already sparse (note the logarithmic color shading), so small sampling differences can lead to noticeable changes in the shape of the upper envelope.

Importantly, the bulk of the distribution (indicated by the darker contours and shading) remains approximately parallel to the zero-surface-humidity bound. This suggests that, in the midlatitudes, variability in T_{500} continues to drive variability in T_s in a manner that scales with the theoretical bound. While variability in q_s could in principle obscure this relationship, it does not appear to do so—likely because its influence is weaker than that of strong free-tropospheric horizontal temperature gradients in the midlatitudes. This fact also motivated the focus of posing a T_{500}-based bound for the midlatitudes: in the tropics and subtropics, T_{500} variability is too small for the bound to be useful, and q_s variability dominates. Therefore, the corrected result (Fig. 2b) remains consistent with the main conclusions of Zhang and Boos (2023).

Closing remarks. To summarize, the zero-humidity bound in Zhang and Boos (2023) is intended as a quantitative reference for the top-down control exerted by a moist-adiabatic mid-to-upper free troposphere, especially in the midlatitudes where variability in T_{500} provides useful guidance on T_s. Specifically: (1) it is a singular limiting case of the adjusted bound with surface moisture also proposed in Zhang and Boos (2023); (2) the inclusion of q^*_{500} corresponds to a dry adiabat connecting the surface to the top of the atmosphere or tropopause, rather than to the 500 hPa level as in Nicolas and Hotz (2025); and (3) the inclusion of q^*_{500} in both bounds is a deliberate choice that balances parsimony and realism. Both the zero-humidity bound and the adjusted bound aim to provide a quantitative reference for the main concept of Zhang and Boos (2023), namely the role of a moist adiabatic mid-to-upper free troposphere in dry heat.

References

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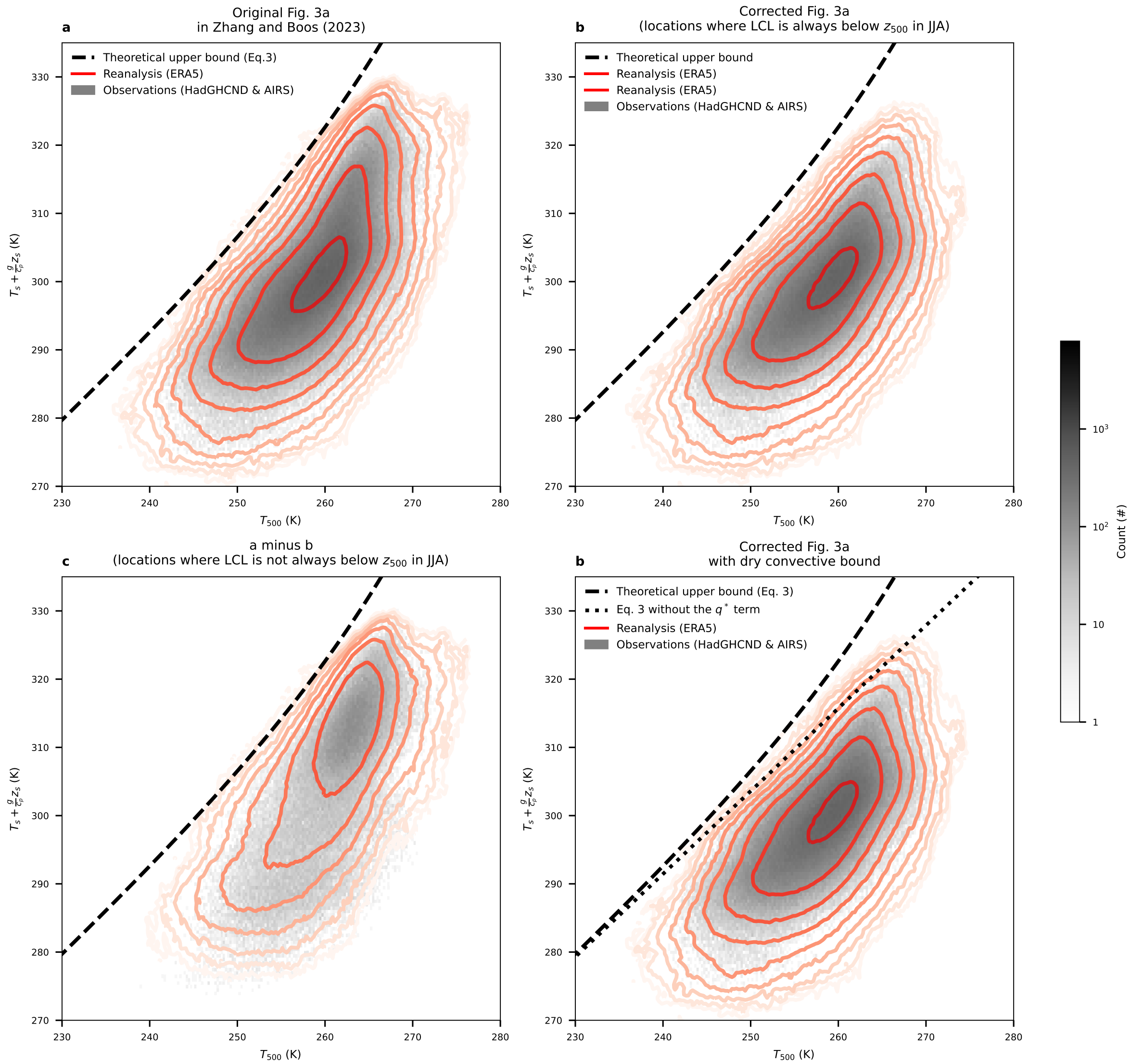

Figure 2: Original Fig. 3a from Zhang and Boos (2023) (a), the corrected version with locations masked where the 500-hPa level does not always reside within the free troposphere in JJA (b), and the difference between the two (c). The corrected Fig. 3a along with the dry adiabatic bound is shown in (d).

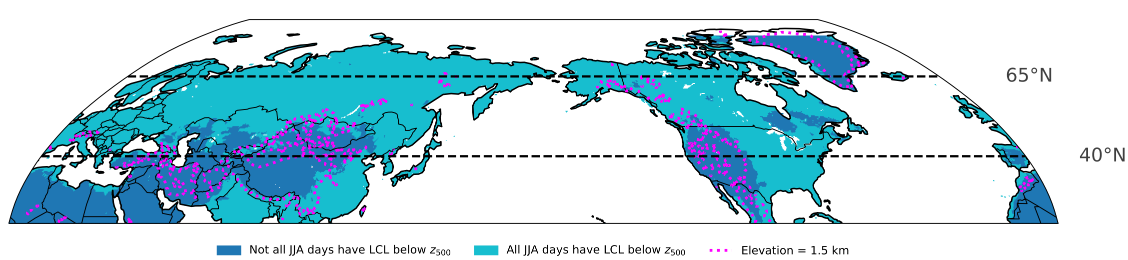

Figure 3: Regions excluded in Fig. 2b (dark blue). Many of these regions, which should have been excluded in the analysis of Zhang and Boos (2023), are over elevated topography (magenta contours), which is the focus of Nicolas and Hotz (2026).